

LAB 4:Subnetting and Supernetting using Cisco Packet Tracer

OBJECTIVE:

- To design and simulate a subnetting and supernetting using **Cisco Packet Tracer**.
- To verify communication between computers using the ping command.

SOFTWARE REQUIRED:

- Cisco Packet Tracer

THOERY:

1. Subnetting:

Subnetting is a technique used in computer networks to divide a large network into smaller subnets. It helps in efficient utilisation of IP addresses, reduces network congestion, and improves security and management. By borrowing bits from the host portion of an IP address, subnetting creates multiple logical networks within a single network address. It is commonly used in organisations to separate departments or segments while using the same base network.

2. Supernetting:

Supernetting is the opposite of subnetting, where multiple smaller networks are combined into a larger network. It is used to reduce the number of routing table entries and simplify routing. Supernetting works by borrowing bits from the network portion of the IP address, allowing several contiguous networks to be represented as a single route. This technique is widely used in large networks and on the Internet to improve routing efficiency and scalability.

NETWORK DESIGN:

Base network 192.168.1.0/24 required number of subnets is 4 and number of IP addresses per subnet=64

Subnet Table:

Subnet	Network address	Broadcast Address	1st usable	Last usable
1	192.168.1.0/26	192.168.1.63/26	192.168.1.1/26	192.168.1.62/26
2	192.168.1.64/26	192.168.1.127/26	192.168.1.65/26	192.168.1.126/26
3	192.168.1.128/26	192.168.1.191/26	192.168.1.129/26	192.168.1.190/26
4	192.168.1.192/26	192.168.1.254/26	192.168.1.193/26	192.168.1.253/26

Supernet Table:

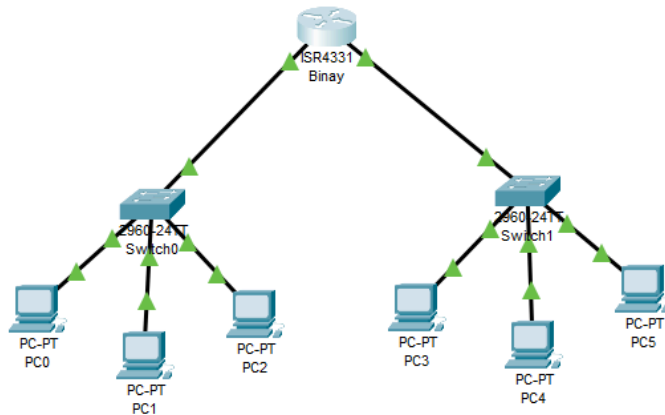
Device	IPV4 Subnetmask	Broadcast Address	Default gateway
Router0	(FastEthernet0/0)	192.168.0.1	255.255.252.0
PC6	192.168.0.10	255.255.252.0	192.168.0.1
PC7	192.168.1.10	255.255.252.0	192.168.0.1
PC8	192.168.2.10	255.255.252.0	192.168.0.1
PC9	192.168.3.10	255.255.252.0	192.168.0.1

NETWORK TOPOLOGY:

Subnetting:

We have two subnets connected to a single router. One consisting PC0,1,2 and another consisting PC3,4,5

Circuit:



OUTPUT:

1. Subnetting:

Step 1: Ping to check if the connection works.

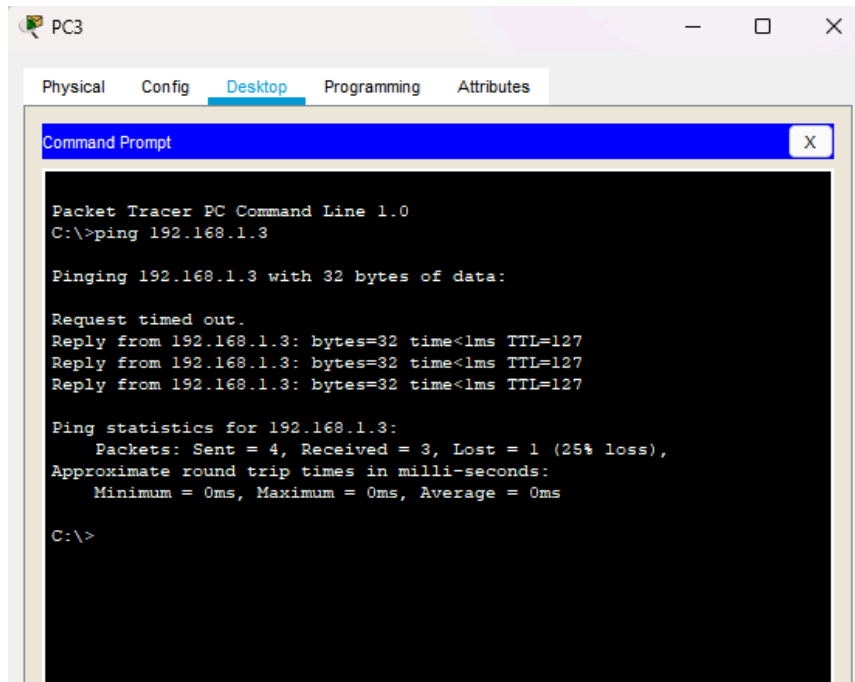
```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.67

Pinging 192.168.1.67 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.67: bytes=32 time<1ms TTL=127
Reply from 192.168.1.67: bytes=32 time<1ms TTL=127
Reply from 192.168.1.67: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.1.67:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

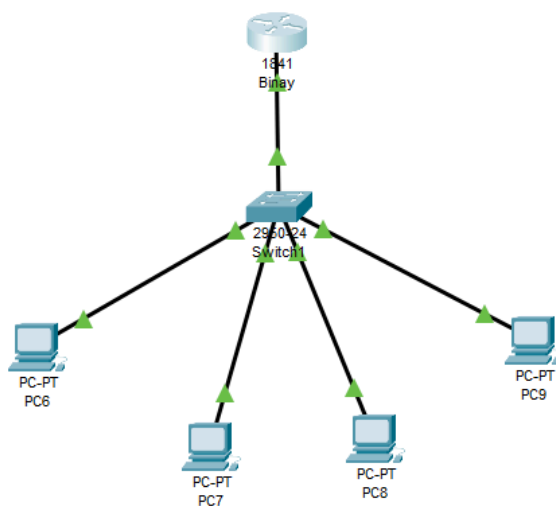
C:\>
```



Supernetting:

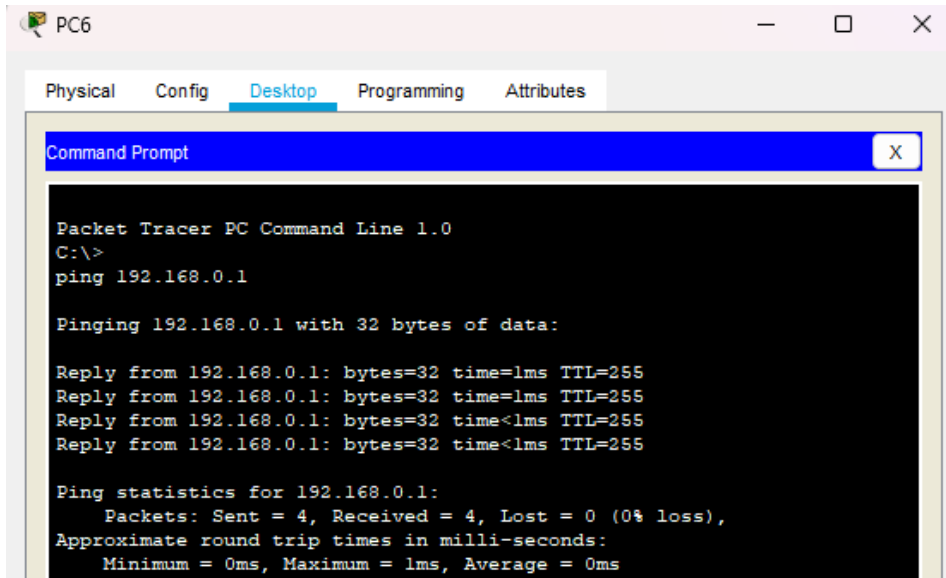
We have single supernet connected to a router . Supernet consisting of PC6,7,8,9 connected to a switch

Circuit:



Supernetting:

Step 1: Ping to check if the connection works



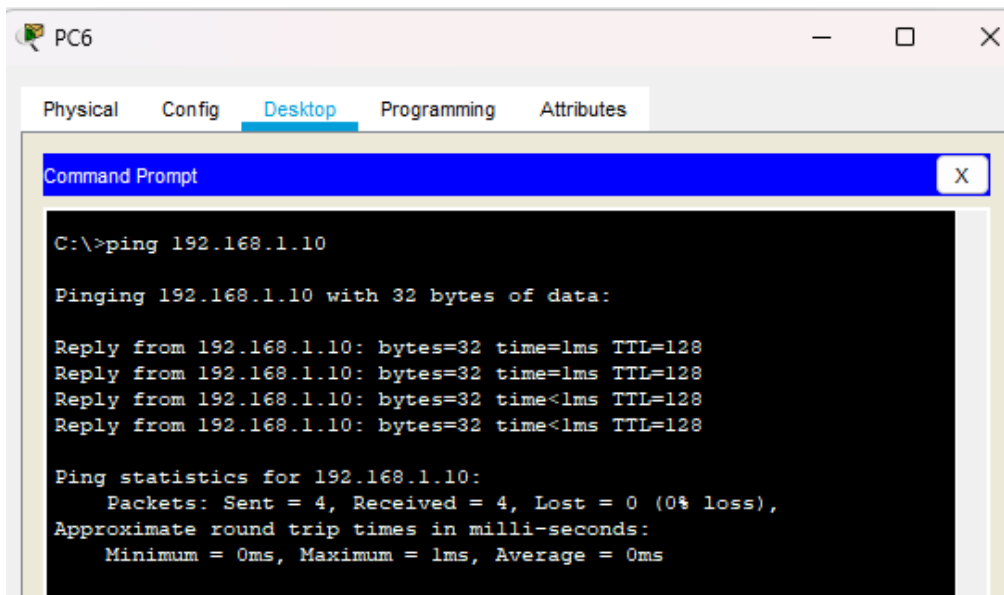
The screenshot shows a Packet Tracer window titled "PC6" with tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, displaying a Command Prompt window. The Command Prompt shows the execution of a ping command to 192.168.0.1, resulting in four successful replies with 0% loss.

```
Packet Tracer PC Command Line 1.0
C:\>
ping 192.168.0.1

Pinging 192.168.0.1 with 32 bytes of data:

Reply from 192.168.0.1: bytes=32 time=1ms TTL=255
Reply from 192.168.0.1: bytes=32 time=1ms TTL=255
Reply from 192.168.0.1: bytes=32 time<1ms TTL=255
Reply from 192.168.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```



The screenshot shows a Packet Tracer window titled "PC6" with tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, displaying a Command Prompt window. The Command Prompt shows the execution of a ping command to 192.168.1.10, resulting in four successful replies with 0% loss.

```
C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time=1ms TTL=128
Reply from 192.168.1.10: bytes=32 time=1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

DISCUSSION:

In this lab experiment, subnetting and supernetting were performed using Cisco Packet Tracer to understand IP address allocation and routing. Subnetting helped to divide a large network into smaller sub-networks, making IP address usage more efficient and reducing network congestion. Supernetting was used to combine multiple networks into a single network, which reduced routing table size and simplified routing. The configurations were tested in the simulator, and successful packet transmission confirmed correct implementation of the concepts.

CONCLUSION:

The subnetting and supernetting lab using Cisco Packet Tracer provided practical knowledge of IP address management and routing efficiency. The experiment reinforced theoretical concepts and demonstrated how proper network design improves performance and scalability. Overall, the lab helped build a strong foundation in designing and managing computer networks.